Spin fluctuation and microscopic ambiguity

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I. Standard spin-fluctuation contraction: exact finite-size identities

For each positive integer `N` let the microscopic algebra be `M_2(C)^{⊗N}` with conventional Pauli matrices. Choose the **declared physical spin product state**

$$\rho_{m,N} = \rho_m^{\otimes N}, \qquad \rho_m = (I + m\sigma_z)/2, \quad 0 < m \le 1.$$

Define collective, centered transverse fluctuations

$$Q_N = (2mN)^{-1/2} \sum_{j=1}^{N} \sigma_x^{(j)}, \qquad P_N = (2mN)^{-1/2} \sum_{j=1}^{N} \sigma_y^{(j)}.$$

Then precisely, **at every finite N**,

$$[Q_N, P_N] = \frac{i}{mN} \sum_{j=1}^{N} \sigma_z^{(j)}, \qquad \langle [Q_N, P_N] \rangle = i.$$

Because independent `sigma_z` variables have variance `1-m²` in the product state,

$$\left\langle ([Q_N, P_N] - iI)^\dagger ([Q_N, P_N] - iI) \right\rangle = \frac{1 - m^2}{Nm^2}.$$

Convergence qualification: this is mean-square convergence *relative to the chosen sequence of states*, not operator-norm convergence. In fact `||[Q_N,P_N]-iI|| = 1+1/m` for every `N` (the all-down eigenvector realizes the maximum), so the operator-norm claim is false even for `m=1`.

For real `u,v` let `a=u/sqrt(2mN)`, `b=v/sqrt(2mN)`. Different sites commute and the single-site spin expectation is obtained by multiplying the two Pauli exponentials. Thus the **exact** finite-N ordered characteristic function is

$$C_N(u, v) = \langle e^{iuQ_N} e^{ivP_N} \rangle = \left(\cos a \cos b - im \sin a \sin b\right)^N.$$

For every fixed `m>0`, `u,v`, taking the logarithm and using the uniform small-angle expansion gives

$$\lim_{N\to\infty} C_N(u, v) = \exp\left[-\frac{u^2 + v^2}{4m} - \frac{iuv}{2}\right].$$

The reversed product has the opposite imaginary Weyl phase. More generally, the analogous single-site expansion for any **finite word** in transverse Weyl exponentials yields the standard Gaussian Weyl algebra, with limiting `[Q,P]=iI` in the regular quasi-free fluctuation representation. This reproduces known quantum central-limit results; it is not a new theorem. Its limiting covariance is `V=(1/(2m)) I_2`, with symplectic matrix `Omega=[[0,1],[-1,0]]`, and `V+iOmega/2 >=0` because `0<m<=1`. The corresponding effective oscillator thermal occupancy is `nbar=(1-m)/(2m)`. This does not mean the original spins were thermal harmonic oscillators.

Numerical controls. The accompanying fluctuation calculation builds full `2^N × 2^N` tensor-product spin matrices for `N=2,3,4,6`; their traces of the ordered exponential products

agree with the factorized exact formula to <2e-13. High-precision direct formula checks at N=2,4,16,64,256,1024,4096 approach the displayed limit. For m=.8,u=v=.7, the limit is approximately 0.7142175684315161 -0.17857058381200364 i and the absolute errors are 0.0024151 at N=16, 0.00014976 at N=256, and 9.36e-6 at N=4096. No numerical evidence is being used as a substitute for the elementary limit proof.

The unpolarized/tracial negative control

At m=0, the preceding Q_N,P_N scaling is undefined. The appropriate unrescaled-variance fluctuations are F_x=N^-1/2 sum sigma_x, F_y=N^-1/2 sum sigma_y in the maximally mixed state rho_0=I/2. They obey

$$\langle e^{iuF_x} e^{ivF_y} \rangle = \left[\cos(u/\sqrt{N})\cos(v/\sqrt{N})\right]^N \longrightarrow e^{-(u^2+v^2)/2},$$

and E([F_x,F_y])=0, E([F_x,F_y]†[F_x,F_y])=4/N. The limit is the characteristic function of **two commuting, independent classical Gaussians**, even though the microscopic operators do not commute. This is a statement about this specific Gaussian fluctuation sector, not a no-go for quantum behavior in all unpolarized systems or other scaling limits.

For a state sequence with m_N→0, the normalized CCR commutator variance is (1-m_N²)/(N m_N²), so it is **not** uniform in polarization. Its decay would require N m_N²→∞, and the oscillator covariance 1/(2m_N) diverges regardless. One cannot extrapolate the fixed-m regular oscillator limit all the way to m=0 using the same normalization.

II. Exact microscopic non-identification under complete collective controls

Compare two distinct microscopic systems initialized in their maximally polarized product states:

- Model A: n spin-1 sites, full space dimension 3^n.
- Model B: 2n spin-1/2 sites, full space dimension 2^(2n).

The initial state in both models is the highest-weight state of **total spin J=n**. Under the total-spin generators J_x,J_y,J_z, both cyclic subspaces carry the unique spin-J=n irreducible representation, of dimension 2n+1. The SU(2) raising/lowering relations on a highest-weight vector uniquely determine all matrix elements within this irrep:

$$J_\pm |J, M\rangle = \sqrt{(J \mp M)(J \pm M + 1)}\, |J, M \pm 1\rangle.$$

Therefore an intertwining unitary between these *cyclic irreducible subspaces* maps the initial vectors and every polynomial and bounded functional calculus of the **collective** J_a to each other. All corresponding collective preparations, collective gates (including nonlinear functions such as J_z²), and collective effects give **exactly equal outcome probabilities**, not merely asymptotically equal ones. This is familiar angular-momentum representation theory, not publication novelty.

For n=3, the complete microscopic Hilbert spaces have dimensions 27 versus 64, but all collective experiments restricted to the common seven-dimensional J=3 irrep agree. A full-matrix implementation checks noncommuting rotation sequences and nonlinear collective evolution for n=1,2,3; discrepancies are below 1e-15.

Scope of equivalence: no claim of equality of full observable algebras. A physically calibrated site-resolved intervention, or operations accessing another total-spin sector, can distinguish the models. The comparison demonstrates that knowing the collective bosonic limit, and even far more than that limit, does not identify microscopic constituent number/type without an adequate observation toolbox.

III. Ricker-filtered emergent bosons do not select geometry

This section deliberately reconnects to the program's *negative controls* only; it does not use a wavelet as a particle ontology. Supply a connected graph with real symmetric positive Laplacian `L`, and define the mathematical scale-response operator

$$W_s = sLe^{-sL}, \quad s > 0.$$

Place `N` independent spins at **each already identified graph vertex** and take the `N→∞` polarized fluctuation limit at fixed `m>0`. It yields an independent canonical pair `Q_i,P_i` at every *assumed* vertex. The filtered fields `Q_s=W_s Q`, `P_t=W_t P` satisfy

$$[Q_s(i), P_t(j)] = i(W_s W_t)_{ij}$$

in the limit. The full 4-site spin matrix calculation verifies the same relation for the **finite-N commutator expectation** without invoking the limit. The scale labels remain redundant/noncanonical; the exact Laplacian zero mode is removed.

An aggregate, mode-blind multiscale commutator measures only

$$\mathcal{A}(s,t) = \frac{1}{V}\frac{1}{i}\sum_i \langle [Q_s(i), P_t(i)]\rangle = \frac{1}{V}\,\mathrm{Tr}(W_s W_t),$$

which depends on the eigenvalues of the supplied `L` alone. In the previous exact graph test, the nonisomorphic 16-vertex rook and Shrikhande graphs have identical Laplacian spectra `0^(1),4^(6),8^(9)`. Both therefore yield

$$\boxed{\mathcal{A}(s,t) = st\left[6e^{-4(s+t)} + 36e^{-8(s+t)}\right]}$$

for every pair of mathematical scales, despite different labeled connectivity. At `s=.12,t=.31`, both values are `0.0829085908049164`, with spectral-formula residual `<5e-17`. The full labeled filtered commutator kernels differ in Frobenius norm by `0.0604811368` under the exhibited fixed vertex labeling.

The latter distinction is physically actionable **only if that vertex identification and spatial measurement protocol have independent meaning**. There is no emergent local measurement basis or spatial metric from CCR or the aggregate wavelet result. Thus bosonic algebra plus multi-scale spectral data does not resolve the already identified inverse-geometry ambiguity. This is a synthesis of known quantum-fluctuation and cospectral-graph phenomena, not a novel theorem.

IV. Microscopic dynamics is a separate ingredient, and bosons do not imply Lorentz invariance

As a controlled physical *example*, supply a one-dimensional periodic spin-1/2 chain with independently calibrated hopping `J>0`, on-site energy `Delta=2J+mu` and an independently specified physical evolution parameter:

$$H = \Delta\sum_j n_j - J\sum_j(\sigma_j^-\sigma_{j+1}^+ + \sigma_j^+\sigma_{j+1}^-), \quad n_j = (I - \sigma_z^{(j)})/2.$$

In the **single-down-spin excitation sector**, the exact restricted Hamiltonian is `Delta I-J A_cycle`, giving

$$E(k) = \mu + 2J[1-\cos(ka)] = \mu + Ja^2k^2 + O(k^4a^4).$$

For `J=.75,mu=.2` on a six-site ring, the independent full `2^6` spin matrix has one-excitation energies `[.2,.95,.95,2.45,2.45,3.2]`, matching the formula to `<9e-16`. At `mu=0`, the one-particle

band touches zero with **quadratic** long-wavelength dispersion (`z=2`), not the `z=1` linear spectrum of a relativistic massless scalar. Calling this toy band's touching point a Goldstone boson would require a separate symmetry-breaking argument; none is asserted for the tuned XY model.

In actual symmetry-broken nonrelativistic many-body theories, Watanabe and Murayama (2012, 2014) show that pairs of broken generators with nonzero commutator expectation can describe a single low-energy mode, often with nonrelativistic dispersion. This is structurally suggestive of the same spin commutator mechanism, but **our product-state central limit is not by itself a spontaneous-symmetry-breaking theorem**.

Ricker filtering `W_s` on such an already-defined spatial generator only reweights wave numbers; it neither changes `E(k)` nor supplies the physical space, time, sources or causality needed for a field theory.